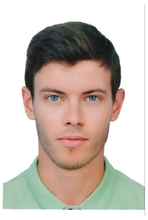

Mohamed KHENAFIF

Embedded Software Engineer | C++ | Real-Time Systems

Massy (91300)

khenafifmohamed@gmail.com

07 57 07 96 52

https://www.linkedin.com/in/mohamed-khenafif-52844335b/

28 ans

B

France

Embedded Software Engineer with deep expertise in bare-metal C/C++ programming and real-time system design on ARM Cortex-M architectures. Passionate about the entire hardware-software lifecycle, from initial silicon bring-up and low-level driver development to performance optimization and complex protocol implementation. Proven ability to manage interrupts, timers, and memory constraints in resource-constrained environments. Seeking to leverage strong debugging and analytical skills in a challen

COMPÉTENCES TECHNIQUES & TRANSVERSALES

- **Programming Languages:** C++, C, Assembly, Python, Shell Scripting.
- **Protocols & Interfaces:** Ethernet, TCP/IP, CAN, SPI, I2C, UART, MQTT (TLS, QoS).
- **Operating Systems:** Embedded Linux, FreeRTOS, Zephyr, Bare-metal.
- **Hardware & Architectures:** ARM Cortex-M, STM32, ESP32, PIC, SoC, Wireless Modules, Sensors
- **Low-Level Expertise:** Hardware Bring-Up, JTAG/RTT Debugging, Interrupt Handling, Timer Configuration, Register Manipulation, Memory Optimization, Performance Tuning
- **Tools & Methodology:** VS Code, PlatformIO, STM32CubeMX, Git, CMake, GitHub Actions

EXPÉRIENCES PROFESSIONNELLES

Freelance Embedded Software Engineer — Depuis décembre 2024 **(1 an)**

Freelance Paris, France

- Architected and developed robust C/C++ firmware for 32-bit microcontrollers (ESP32, ARM Cortex-M) in resource-constrained environments, ensuring high performance and reliability.
- Performed complete hardware bring-up on custom PCBs, validating core system functionality and peripheral interfaces from scratch.
- Designed and implemented software architectures using FreeRTOS, creating and managing multitasking systems with queues, semaphores, and software timers for real-time applications.
- Developed low-level drivers for sensors and actuators, integrating communication protocols (UART, I2C, SPI, Ethernet, BLE) with a focus on data integrity and performance.
- Utilized JTAG and RTT for in-depth debugging of complex hardware-software interactions, identifying and resolving issues at the register level.

Data Automation Engineer (Python) Intern — De mars 2024 à septembre 2024 **(6 mois)**

Rhon TELECOM Paris, France

Developed Python scripts to automate the processing and analysis of large datasets for FTTH network infrastructures, improving operational efficiency.

Embedded Software Engineer — De mai 2022 à mai 2023 **(1 an)**

IS PESAGE Algiers

Participated in the full development lifecycle, from technical specifications to functional testing and validation on embedded targets, ensuring system reliability and precision under operational conditions.

Embedded Software Engineer Intern — De février 2021 à août 2021 **(6 mois)**

SONATRACH Hassi R'mel

Designed and developed an autonomous real-time embedded system featuring a closed control loop integrating sensors and actuators.

FORMATIONS ET CERTIFICATIONS

Master 2 – Optical Networks and Photonic Systems — De 2023 à 2024

Institut Polytechnique Paris Paris

State Engineer in Electronics — De 2016 à 2021

Ecole Nationale Polytechnique Alger

Specialization: Embedded Systems and Real-Time Software.

C Language Training — 2019

CISCO Networking hardware company

Linux Training — 2018

CISCO Networking hardware company

LANGUAGES

English: Fluent ||||| **French:** Fluent ||||| **Arabic:** Native

SOFT SKILLS

Autonomous, Rigorous, Curious, Analytical Mindset, Adaptable, Sense of Responsibility, Team Player, Priority Management.